

AI Engineering Learning Program

16 classes, structured into daily 1–2h sessions. Each day includes study content, practice tasks, resources, and knowledge-check questions.

48

STUDY DAYS

8

MODULES

~90

HOURS TOTAL

0

COMPLETED

Overall Progress

0%

● Study ● Practice ● Resources ● Knowledge Check

WEEK 1

FOUNDATIONS

01

WEEK 1

4h



Intro to AI Engineering

DAY 1

What is AI Engineering & The Stack

🕒 1.5h

▶ STUDY TOPICS

- AI Engineering vs classical ML: key distinctions (data-centric vs model-centric)
- The modern AI engineering stack: LLM providers, orchestration, vector DBs, observability
- Role breakdown: what AI engineers actually build vs ML researchers

◆ PRACTICE TASK

Write a 300-word comparison: when would you use a fine-tuned model vs a foundation model + prompt engineering? Use your ServiceNow chatbot as context.

○ RESOURCES

↗ What is AI Engineering (Chip Huyen) ↗ The AI Engineer (Swyx) ↗ Andrej Karpathy: Software 2.0

+ KNOWLEDGE CHECK

Which best describes the AI Engineer's primary work?

A. Training neural networks from scratch on custom datasets

B. Composing foundation models, APIs, and tooling to build production systems

C. Writing CUDA kernels for GPU optimization

D. Creating ML research papers and benchmarks

DAY 2

Practical Use Cases & Business Fit

🕒 1.5h

STUDY TOPICS

- LLM use case taxonomy: text generation, classification, extraction, Q&A, agents
- Decision framework: LLM vs fine-tune vs classical ML vs rule-based
- Cost, latency, accuracy trade-offs — the production triangle

PRACTICE TASK

Map 3 real use cases from your Disney Streaming / ServiceNow experience to the decision framework. Which LLM pattern fits each and why?

RESOURCES

↗ When NOT to use LLMs (Hamel Husain) ↗ OpenAI Cookbook — Techniques ↗ Building LLM Applications for Production

KNOWLEDGE CHECK

A compliance team needs to classify 10,000 legal docs/day into 5 fixed categories. Best approach?

A. GPT-4 with zero-shot classification for each document

B. Fine-tuned smaller model (e.g., BERT/DistilBERT) with labeled data

C. Agentic RAG with tool calling

D. Prompt chain with CoT reasoning

WEEK 2

FOUNDATIONS

02

WEEK 2

Foundational Models

6h



WEEK 3

CORE SKILLS

03

WEEK 3

Prompt Engineering

6h



WEEK 4

QUALITY & MEASUREMENT

04

WEEK 4

Evaluations for AI Systems

6h

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WEEKS 5-6

KNOWLEDGE SYSTEMS

05

WEEKS 5-6

Embeddings, Vectorization & RAG

12h

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WEEKS 7-8

AGENTS

06

WEEKS 7-8

Building AI Agents

10h

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WEEK 9

DEVEX & PRODUCTION

07

WEEK 9

DevEx + Production Readiness

8h

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WEEK 10

CAPSTONE

08

WEEK 10

Final Project & Portfolio

8h

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